Question 2 - Isobar currents at (72°)

The dynamical-suppression argument is well taken. Could the proponents provide an order-of-magnitude estimate of the residual isobar-current contribution after dynamical suppression at $Q^2 = 4 - 5$ GeV^2, so that the PAC can assess whether it constitutes a significant and quantifiable systematic uncertainty for the (72°) setting?

As the figure from Ref. https://inspirehep.net/literature/360093) shows, the $\gamma^* N \to \Delta$ transition form-factor is suppressed almost by factor of 2 as compared to the elastic $\gamma^* N \to N$ form factor. This results in a factor of 4 in the suppression of the re-scattering cross section as compared to diagonal $NN \to NN$ re-scattering contribution. Note that since $N\Delta \to NN$ re-scattering amplitude does not interfere with diagonal $NN \to NN$ amplitude, the former is predominantly real (because of pion exchange nature of scattering) while the latter is imaginary because of dominance of the Pomeron exchange at these energies.

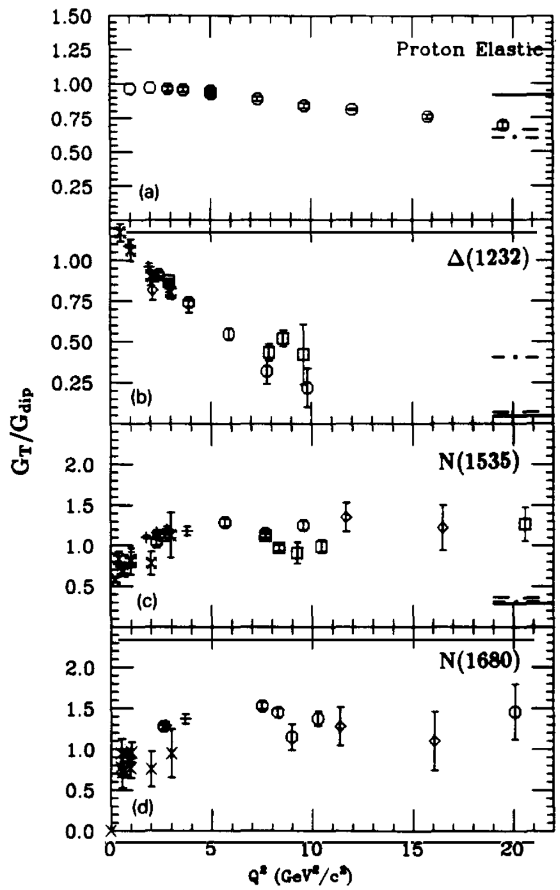

Fig. 19. (a) The absolute value of the proton elastic magnetic form-factor divided by the dipole shape, G_M/G_{dipole}, versus Q^2, where $G_{\text{dipole}} = \mu_P(1 + Q^2/0.71)^{-2}$. The data are from ref. [Ar-86]. The curves at the right of the figure are the results of calculations [Ji-87] at $Q^2 = 20\,\text{GeV}^2/c^2$ using the proton distribution functions C-Z (solid), G-S (dot–dashed), and K-S (dashed). (b)–(d) The quantity G_T/G_{dipole} versus Q^2 for transitions to the first, second and third resonances, respectively, with $G_{\text{dipole}} = 3(1 + Q^2/0.71)^{-2}$. The first resonance (b) is the Δ(1232). The second resonance (c) at $Q^2 = 0$ consists of contributions from the $D_{13}(1520)$ and $S_{12}(1535)$, but at $Q^2 \sim 3\,\text{GeV}^2/c^2$ it is mainly due to the $S_{12}(1535)$. The third resonance (d) at low Q^2 is dominated by the $F_{15}(1680)$. The resonance form-factor G_T is defined in the text (eq. 9). The fits for G_T were based on inclusive data reconstructed from refs. [Br-76] (+), [Ro-92] (○), [Po-74, Bo-79, Ri-73] (□, ◇). Also shown at lower Q^2 denoted by (×) are form-factors derived from amplitudes obtained from exclusive (e, e′p)π° and (e, e′p)η data obtained from refs. [Br-84, PDG-92, Bu-91]. The errors shown are statistical. Estimated systematic errors are discussed in the text.

It is worth nothing that the Δ form factor is larger than elastic form factor at $Q^2 < 0.5\ \mathrm{GeV}^2$ which is why intermediate Δ contribution is important for low energy electro-disintegration processes (e.g. https://inspirehep.net/literature/134355).

In addition, there is a kinematical suppression, due to the fact that the Δ production threshold is at

$$x_\Delta = \frac{1}{1 + \frac{m_\Delta^2 - m_N^2}{Q^2}}$$

while the peak of the FSI takes place at x = 1. As a result the deuteron wave function in the case of intermediate Δ production has an additional longitudinal component of the momentum $\approx m_N(x - x_\Delta)$ for $x > x_\Delta$